

⇒ Trigonometric Substitutions:-

$\sqrt{a^2 - x^2} \rightarrow x = a \sin \theta$	$\cos^2 \theta = 1 - \sin^2 \theta$
$\sqrt{a^2 + x^2} \rightarrow x = a \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 + a^2} \rightarrow x = a \sec \theta$	$\tan^2 \theta = \sec^2 \theta - 1$

→ Note: in case (i) it is not compulsory that we will just substitute in the square root case. or its power not compulsory always square it can come in upon form.

ex:- 1

$\int_0^{\frac{3}{2}} \frac{dx}{\sqrt{9 - x^2}} \rightarrow$ it matches with the first formula.

$\int_0^{\frac{3}{2}} \frac{dx}{\sqrt{3^2 - x^2}} \quad \text{so } x = a \sin \theta$
 $x = 3 \sin \theta$

Now we have to shift x with θ
 so take the derivative of $x = 3 \sin \theta$.

$$[dx = 3 \cos \theta d\theta]$$

Now limits put in $x = 3 \sin \theta$

$$\frac{3}{2} = 3 \sin \theta$$

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Applying $\sin^{-1}$ on both sides we will get.

$$\theta = \sin^{-1}\left(\frac{1}{2}\right)$$

which is $= \frac{\pi}{6}$ in Radian.

Now for lower limit which is Zero.

$$r = 3 \sin \theta$$

$$0 = 3 \sin \theta$$

$$0 = 3 \sin \theta$$

$$\sin \theta = 0$$

if we find the value of ~~sin~~ θ we will get

$$\theta = \sin^{-1}(0)$$

$$\text{and } \sin 0 = 0.$$

$$\text{so } \theta = 0.$$

= so lower limit = 0 and upper = $\frac{\pi}{6}$.

$$\int_0^{\frac{\pi}{6}} \frac{3 \cos \theta \, d\theta}{\sqrt{3^2 - (3 \sin \theta)^2}}$$

$$\int_0^{\frac{\pi}{6}} \frac{3 \cos \theta \, d\theta}{\sqrt{3^2 (1 - \sin^2 \theta)}}$$

$$= \int_0^{\frac{\pi}{6}} \frac{\cos \theta}{\sqrt{\cos \theta}} d\theta$$

note. $1 - \sin^2 \theta = \cos^2 \theta$

$$\Rightarrow \int_0^{\frac{\pi}{6}} (1) d\theta$$

$$= \left[\theta \right]_0^{\frac{\pi}{6}} = \frac{\pi}{6} - 0 = \frac{\pi}{6} \text{ Ans}$$

upper limit lower limit

$$= \tan^2 \theta$$

$$\int \tan^2 \theta$$

$$= \int \tan \theta$$

$$= \int \theta$$

$$= \int$$